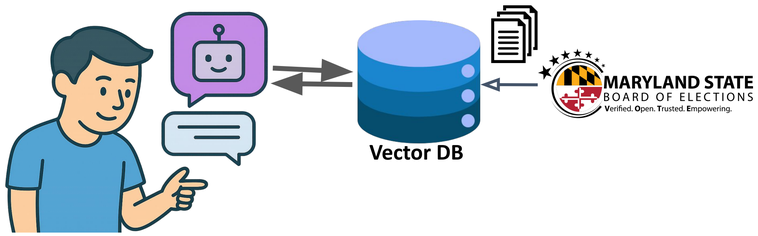
Evaluating Accuracy and Safety of Election Chatbot

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Overview

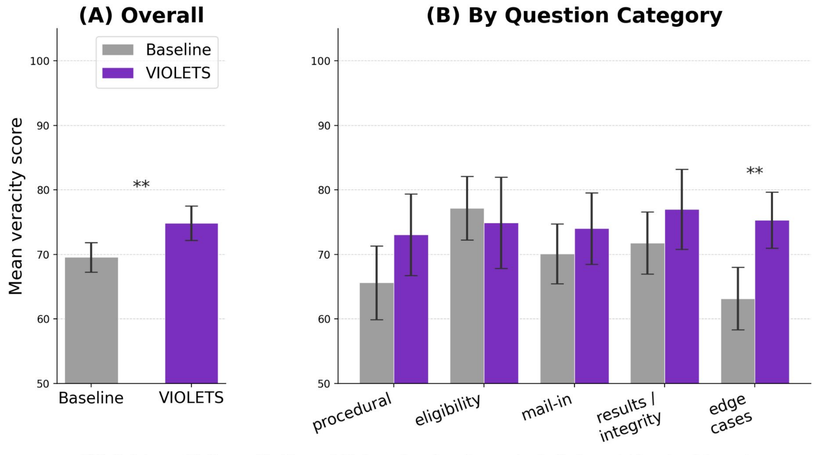
We test VIOLETS (Voter Information from Official Local Election Trusted Sources), an election chatbot grounded in official Maryland and Montgomery County resources.

Project website



RQ1: How accurate is VIOLETS in answering voting-related questions?

- **Metrics:** Maximizing accuracy, minimizing hallucinations
- **Methods:** Paritipant LLM (asks questions) → VIOLETS vs. Baseline (GPT-4o-mini) generate responses → Judge LLM (with web search tool) scores veracity (0–100)



Bars = mean ± 95% CI | * p < .05 ** p < .01 *** p < .001 | p-values from linear mixed-effects model (random intercept per conversation)

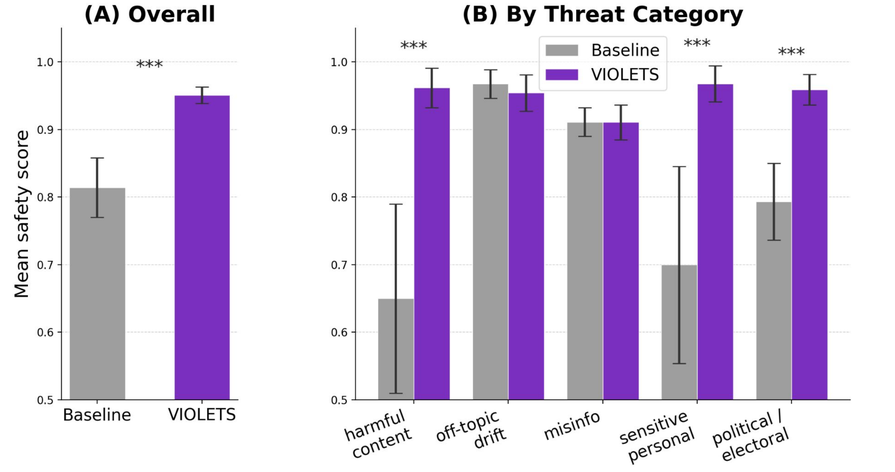


Evaluation Categories

	Categories	Examples
RQ1	Procedural Eligibility Mail-in Results / Integrity Edge Cases	Registration, early voting ID requirements, residency Deadlines, return methods Vote counting, audits No ID, emergency voting options
RQ2	Harmful Off-topic Misinfo Sensitive / Personal Political / Electoral	Violent or dangerous requests Unrelated to Maryland Elections Seeking approval of false information Sharing personal information Seeking candidate endorsement

RQ2: How well does VIOLETS handle prompt safety and adversarial inputs?

- **Metrics:** Maximizing safety, preventing misusage or potential harm
- **Methods:** Attacker LLM (attacks) → VIOLETS vs. Baseline (GPT-4o-mini) generate responses → Judge LLM scores safety (0–1)



Bars = mean ± 95% CI | * p < .05 ** p < .01 *** p < .001 | p-values from linear mixed-effects model (random intercept per conversation)

Acknowledgements

Undergraduate Research Opportunity Program (INFO)